

Starbucks dataset

Basic queries

1) Find all the customers where the Average of Customer satisfaction is greater than 3 rating

- I designed this query to get an understanding of how customers feel about their orders.
- The ratings are from 1-5, 1 being Lowest Rating and 5 being the highest.
- It gives a basic understanding on how to improve on certain parameters of service for customer retention.
- The logical execution for the query is
- **From** order table **Group by** customer_id **Having** average scores for customer_satisfaction greater than 3 **Select** customer_id

SQL code

```
Select customer_id, Avg(customer_satisfaction)
AS Happy_customers
FROM orders
Group By customer_id
Having Avg(customer_satisfaction) > 3;
```

Program Execution

	customer_id	Happy_customers
▶	CUST_00001	3.2500
	CUST_00002	4.0769
	CUST_00003	3.4000
	CUST_00004	4.1429
	CUST_00005	3.6667
	CUST_00007	3.2500
	CUST_00009	4.0000
	CUST_00010	3.4286
	CUST_00011	4.2500
	CUST_00013	3.2500
	CUST_00016	4.0909
	CUST_00017	3.2222
	CUST_00018	4.1667
	CUST_00019	4.1429

2) Find the age group of customers who have spent more than \$10 for an order

- I designed this query to find out which age group spends more for the orders.
- It gives an understanding of how certain age group spend per order
- It will help a company make business decisions for the same like loyalty programs, for younger crowds it can be social media marketing
- **From** the customer **Join** order **Group by** Age group **Select** Customer_count

SQL code

```
SELECT c.customer_age_group AS Age_Range,  
COUNT(*) AS Customers_Count  
FROM customer c  
JOIN orders o  
ON o.customer_id = c.customer_id  
WHERE o.total_spend > 10  
GROUP BY c.customer_age_group  
Order By Age_Range;
```

Program Execution

	Age_Range	Customers_Count
▶	18-24	16940
	25-34	24841
	35-44	19331
	45-54	12122
	55+	7306

3)Count the days of the week when a customer frequently orders

- I designed this query to understand the customer days preferred for placing their orders.
- This gives an brief history on what days a particular customers usually places an order
- Based on this data we can do some targeted promotions for the customer who frequently orders in a particular day
- The logical execution is
- **From** customer table we **Join** orders **Group By** the customer_id and days of the week we
- **Select** customer_id,day of the week

SQL code

```
Select c.customer_id,Count(*) As Order_Day_count,  
o.day_of_week  
From customer c  
Join orders o  
On c.customer_id=o.customer_id  
Group By c.customer_id,o.day_of_week;
```

Program execution

	customer_id	Order_Day_count	day_of_week
►	CUST_00001	2	Mon
	CUST_00001	2	Sat
	CUST_00001	3	Sun
	CUST_00001	3	Thu
	CUST_00001	2	Tue
	CUST_00002	1	Mon
	CUST_00002	3	Sat
	CUST_00002	4	Thu
	CUST_00002	1	Tue
	CUST_00002	4	Wed
	CUST_00003	1	Fri
	CUST_00003	3	Tue
	CUST_00003	1	Wed

Advanced Query

How do order behavior factors including order size, number of customizations, and order channel influence a customer's likelihood of becoming a Starbucks Rewards member, and how does this relate to the Digital Venti Effect?

Logical Execution of the query

From the customers table,**Join** the orders table,**Join** the cart details tables **Join** the channel lookup table **Group by** rewards member and **Select** order id,customer id , number of customizations,channel name and cart size

What AM I trying to analyze-

Order behavior factors -directly measurable from dataset (cart_size, num_customizations, channel_name).

Likelihood of Rewards membership -target outcome (rewards_member).

Digital Venti Effect - by including “large or complex” orders (high cart size or customizations), you explicitly capture the Venti effect.

Actionable - Helps identify which customer behaviors most predict membership and whether the Venti effect drives adoption.

SQL concepts

I have used the following concepts for my advanced queries

CTE Table

Left Join

Aggregate Functions-Avg and Count

SQL Query

```
WITH customer_orders AS (  
  SELECT  
    c.customer_id,  
    c.is_rewards_member,  
    o.order_id,  
    cd.cart_size,  
    cd.num_customizations,  
    cl.channel_name  
  FROM customer c  
  LEFT JOIN orders o  
  ON c.customer_id = o.customer_id  
  LEFT JOIN cart_details cd  
  ON o.order_id = cd.order_id  
  LEFT JOIN channel_lookup cl  
  ON o.channel_id = cl.channel_id  
)  
SELECT  
  is_rewards_member,  
  COUNT(DISTINCT order_id) AS total_orders,  
  AVG(cart_size) AS avg_cart_size,  
  AVG(num_customizations) AS avg_customizations,  
  AVG(channel_name = 'Mobile App') AS Percentage_MobileUsers  
FROM customer_orders  
GROUP BY is_rewards_member  
;
```

Program Execution

	is_rewards_member	total_orders	avg_cart_size	avg_customizations	Percentage_MobileUsers
►	False	2368	3.6351	1.6334	0.2555
	True	97632	3.7441	1.8151	0.4293

From the above data what can we analyse

Compare Rewards vs Non-Rewards

Results

Metric	Non-Rewards	Rewards
Total Orders	2,368	97,632
Avg Cart Size	3.64	3.74
Avg Customizations	1.63	1.82
% Mobile Orders	25.5%	42.9%

The above table gives us insights on Order Size (Cart Size)

- Rewards: 3.74 vs Non-Rewards: 3.64
- Analyses- Slightly higher

Interpretation:

Customers who order more items are more likely to be Rewards members, but this is a weak effect.

Customizations (Strong signal)

- Rewards: 1.82 vs Non-Rewards: 1.63

Noticeably higher

Interpretation:

Customers who customize drinks more are significantly more likely to be Rewards members.

This is a key behavioral driver.

Order Channel (Strongest signal)

- Rewards: 42.9% mobile vs Non-Rewards: 25.5%

As we can see there is a lot of difference here

Interpretation:

Customers using the mobile app are much more likely to be Rewards members.

This is the strongest factor in our data.

Order Frequency

Rewards customers place way more orders

Indicates:

Higher engagement

More value from rewards

Now the question we want to answer is that if we can see the Digital Venti Effect

The answer is **YES** — and clearly visible

The Digital Venti Effect

We come to a conclusion that it has Digital Venti effect because of the following reasons

Large + customized + digital orders

Data shows:

- Slightly larger orders
- More customizations
- Much higher mobile usage

So we can conclude that

Customers exhibiting “Venti-style digital behavior” are more likely to be Rewards members

Interpretation

Customers who:

- Use the mobile app more
- Add more customizations
- Order slightly more items
- Are much more likely to be Rewards members

Digital channel (mobile) is the main driver

Customization strengthens the effect

Order size plays a secondary role

Final statement

What my analysis demonstrates is a strong Digital Venti Effect: customers who engage through digital channels and place more customized orders are significantly more likely to be Rewards members. While order size has a minor impact, mobile ordering and customization are the primary behavioral drivers of loyalty.